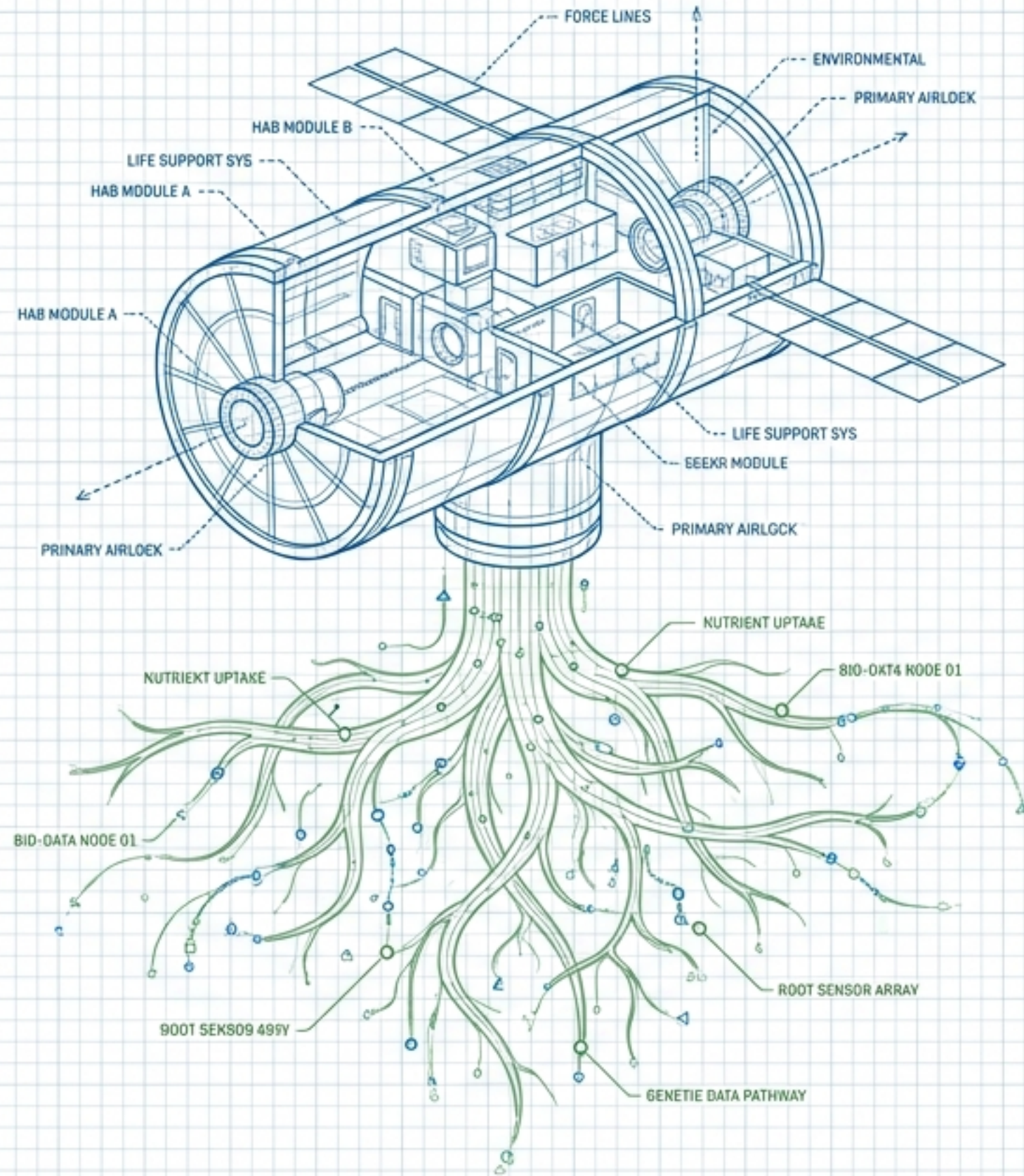




Engineering the Architecture of Off-World Life

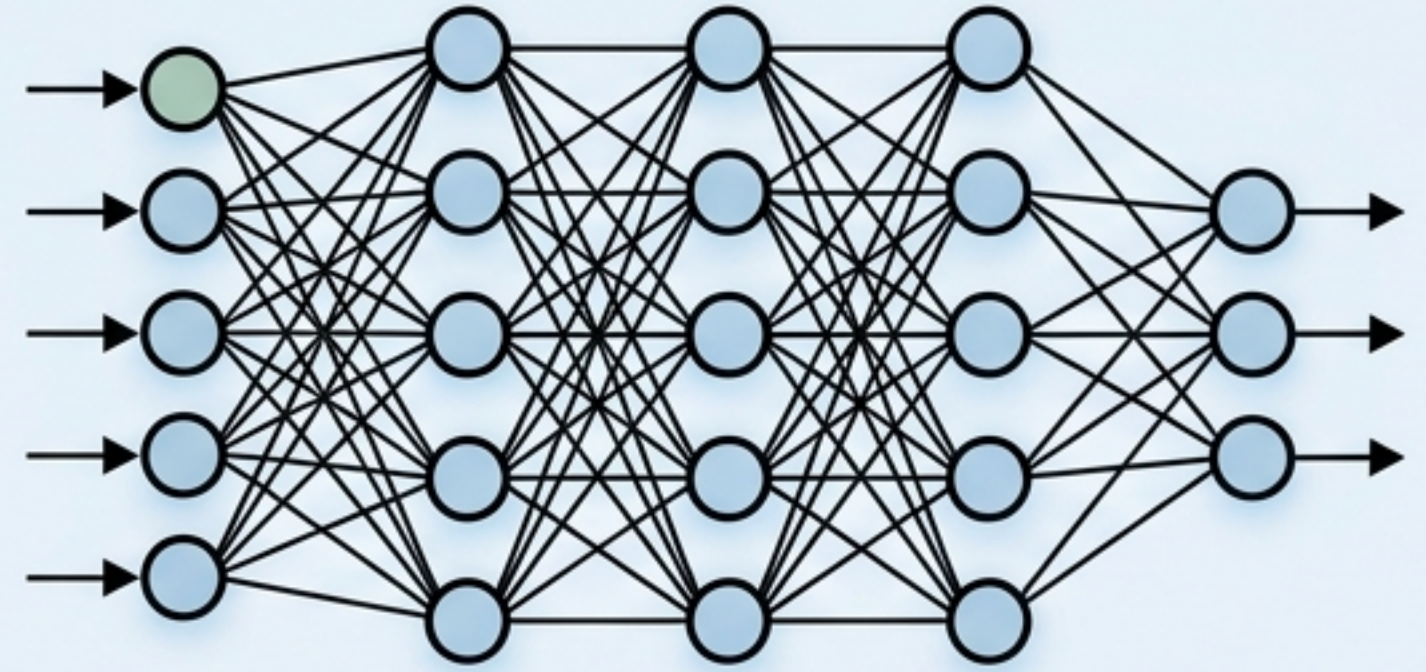
The Collaborative Science Environment (CoSE)
by Dr. Richard Barker

System Status: Online
Total Commits: 997
Primary Data Source:
NASA GeneLab / OSDR Ecosystem



The Scientist

Title	Genetic Engineer
Affiliation	Purdue University / Astrobiology
Focus	Lunar LEAF project, plant responses to the lunar/Martian environment.



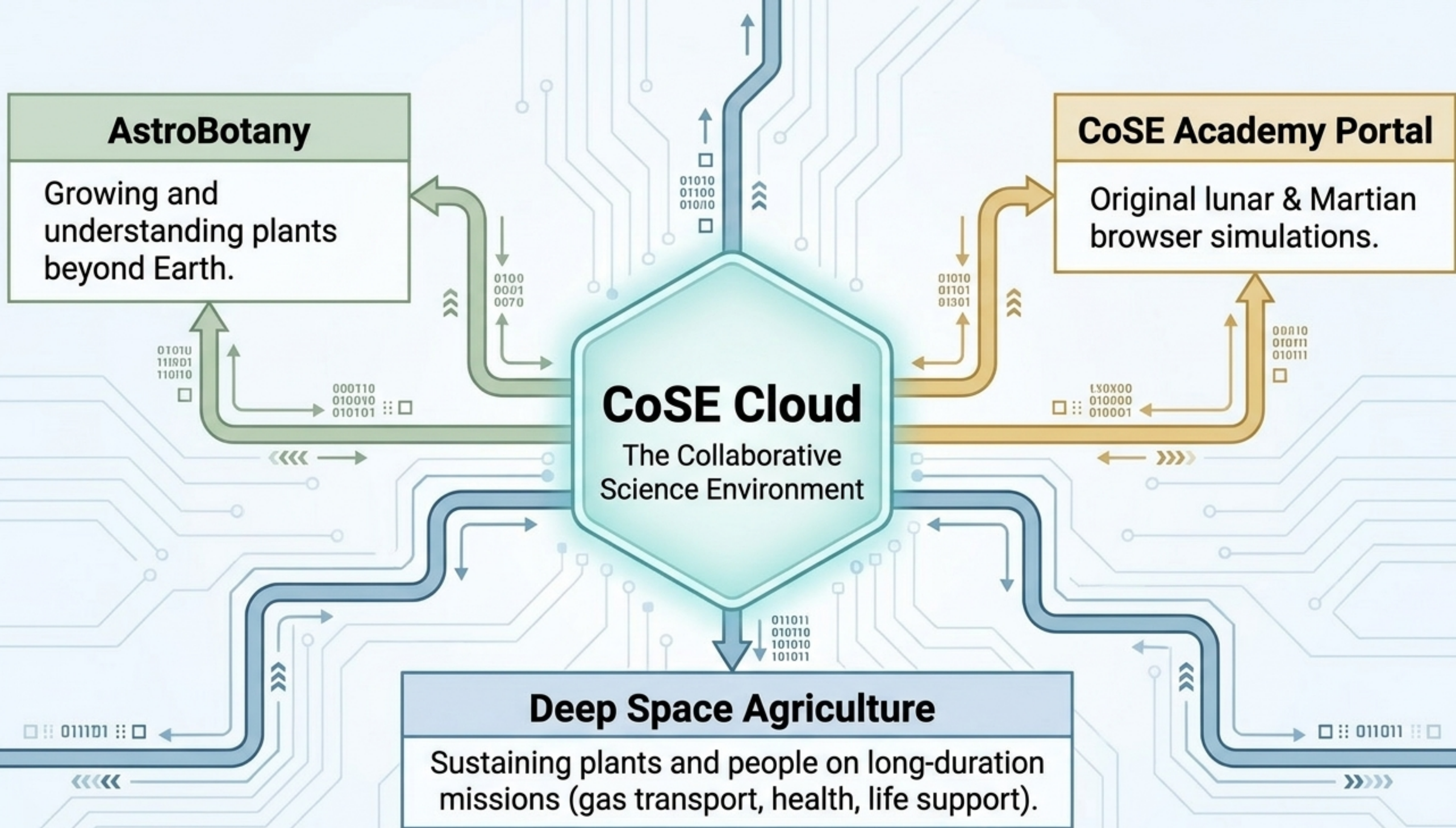
The Technologist

Title	Bioinformatician
Role	Self-taught AI/ML developer & open-source builder.
Focus	Building FAIR, open tools and data processing pipelines for space agriculture.

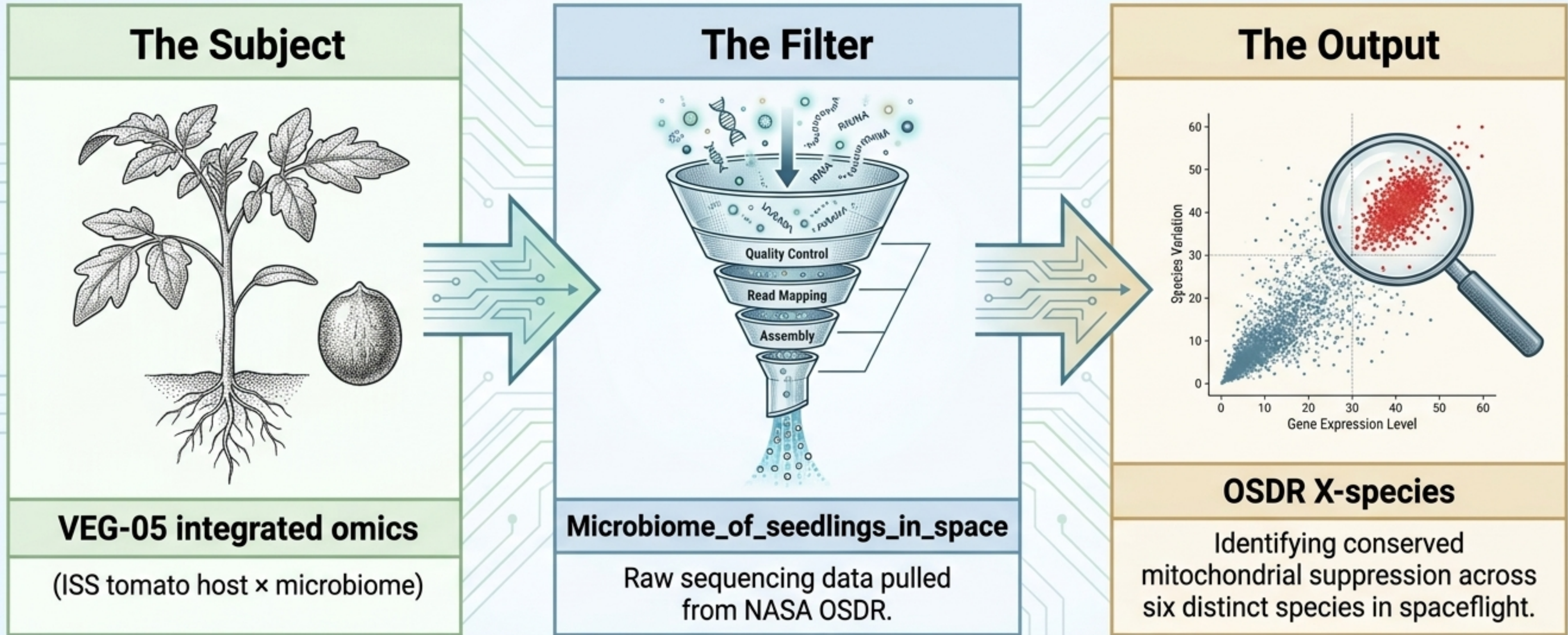
Mission Objective: Building open tools and data for space agriculture.

The Challenge of Deep Space Agriculture

Biological Constraint	Earth Environment	Spaceflight Constraint 	CoSE Target Repository
Gravity & Growth	1G, natural tropism dictates root orientation	Microgravity, disrupted spatial orientation	Tropism_autodecoder_2026
Cellular Stress	Standard atmospheric shielding	Radiation exposure, severe mitochondrial suppression	OSDR_X-species
Survival Resources	Open ecosystem agriculture	Closed-loop life support, human muscle atrophy	VEG-05 integrated omics & Muscle-Atrophy-Multi-Omics



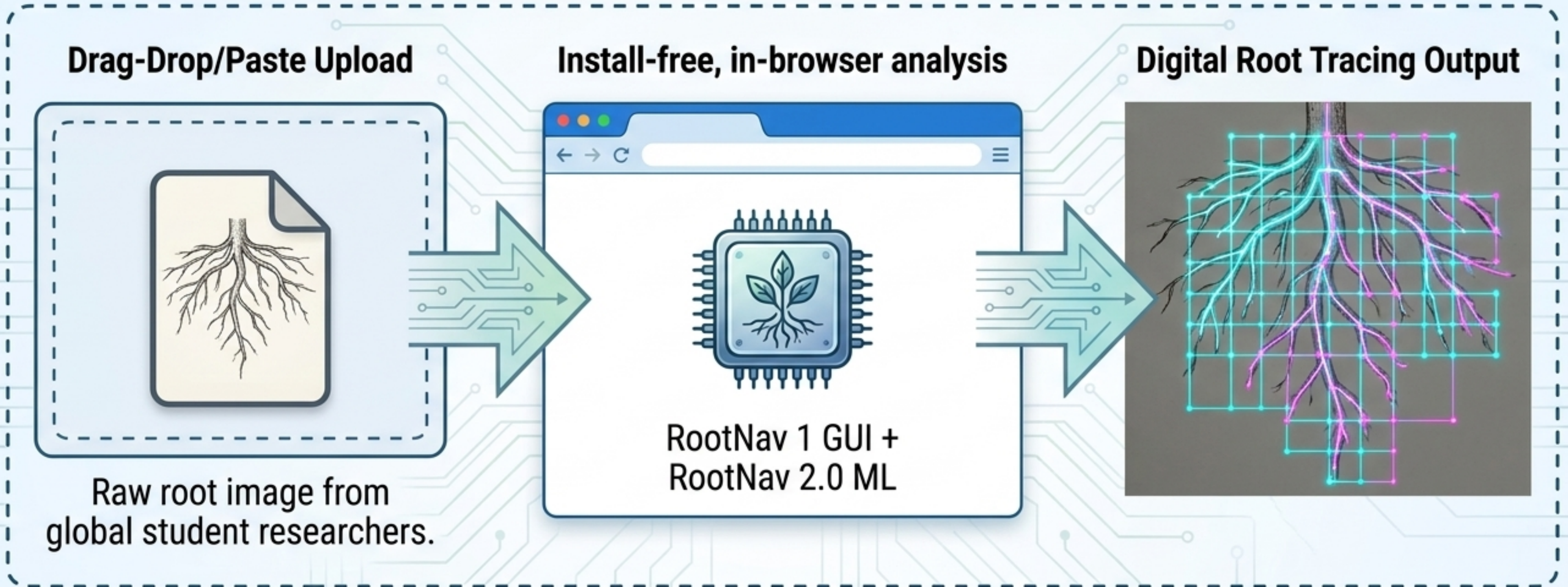
Pillar I: AstroBotany & Multi-Omics



Key Takeaway: Automated pipelines extract FAIR, open-source biological insights from deep space raw data.

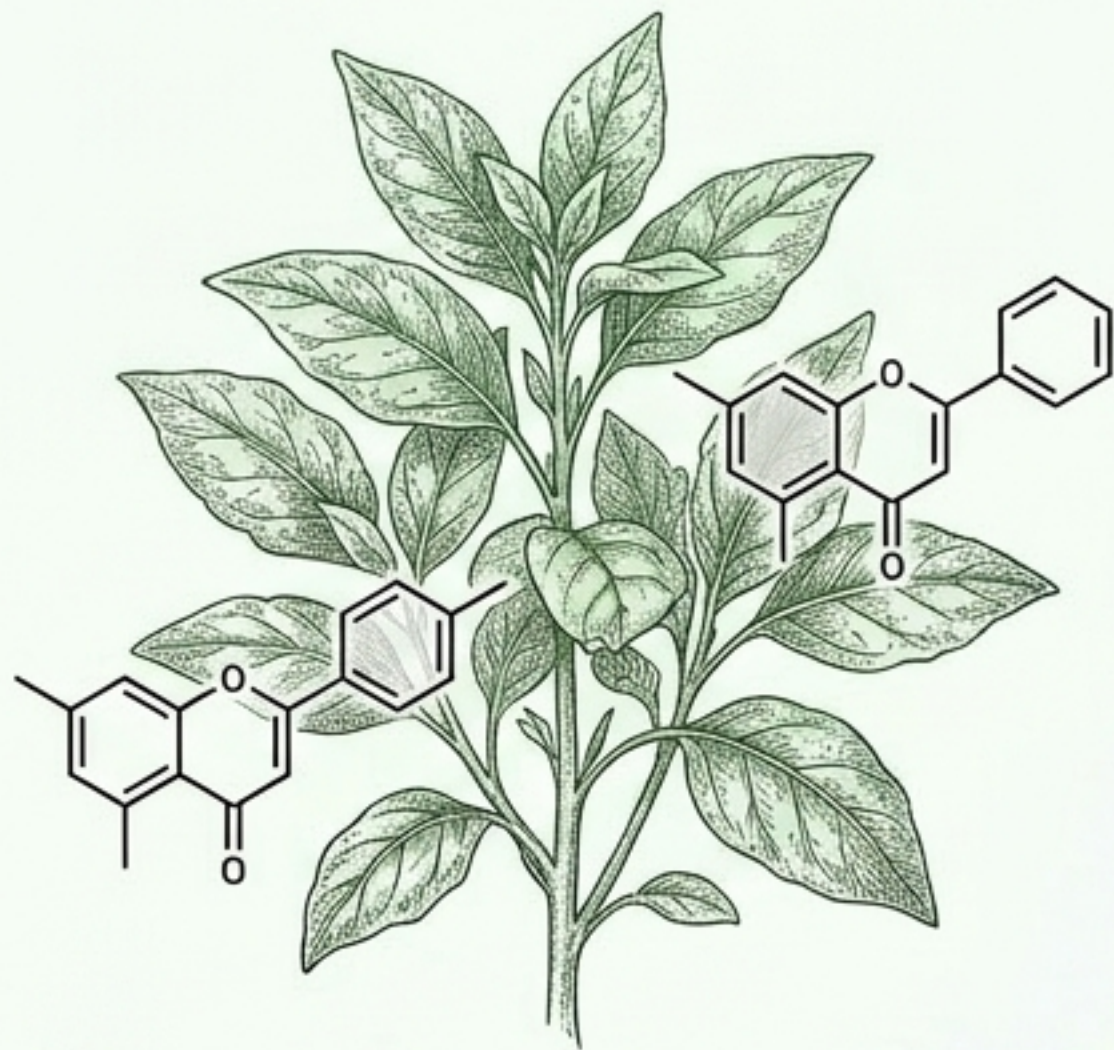
Democratizing Plant Data: The astroroot Engine

Part of AIRI (AstroBotany International Research Initiative)



Zero-install architecture allows students globally to train models and run single or batch analyses directly from the browser.

Botany



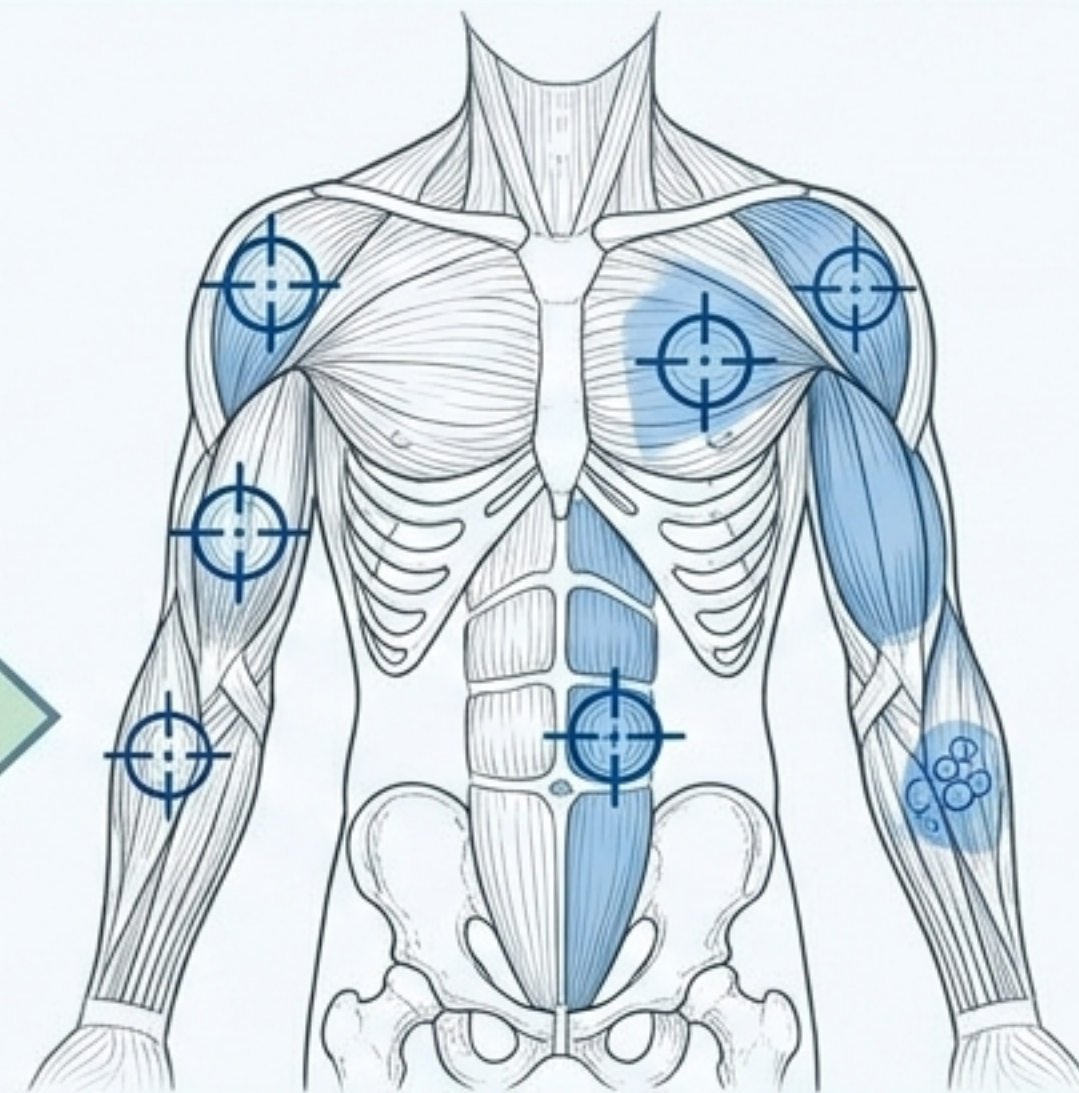
Astronaut_brain_food

Cultivation of food-derived flavonoids in microgravity.



The Countermeasure Pathway

Human Physiology

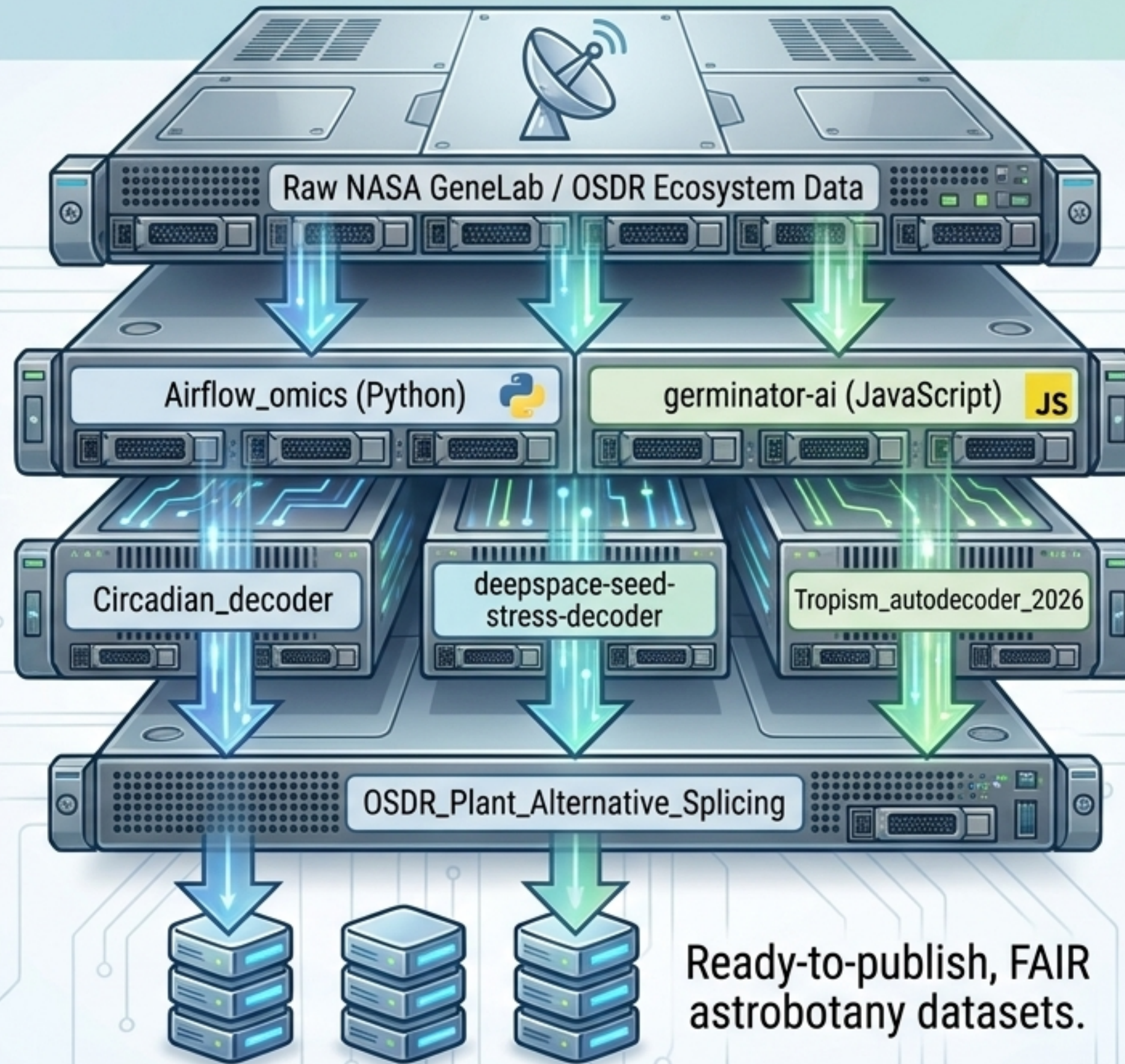


Astronaut Health Search & Muscle-Atrophy-Multi-Omics

Tracking spaceflight oncogenic biomarkers and muscle degradation.

Key Insight: Discovering and cataloging the exact relationship between spaceflight oncogenic biomarkers and their reversal through targeted, food-derived flavonoids.

The Bioinformatics Engine Room



CoSE Academy Portal: Simulating the Frontier

Settlers_of_the_Moon_or_Mars

3D models for settlers board game integrated with Jupyter Notebooks for data-driven gameplay.

Lunar_OpenSC2K

A JavaScript-based lunar settlement simulator inspired by classic Maxis sims.

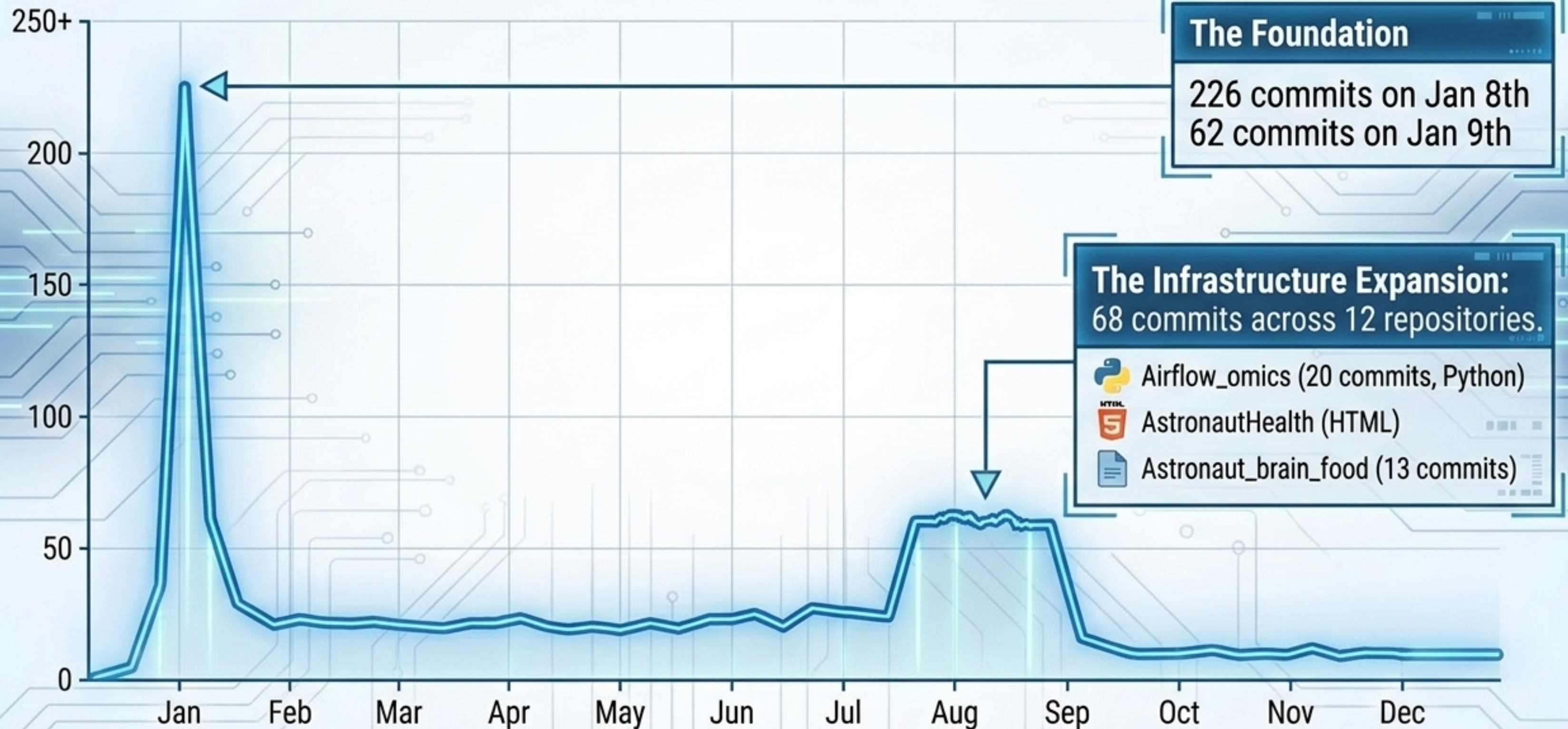
Lunar_OpenSC2K

A JavaScript-based lunar settlement simulator inspired by classic Maxis sims.

Translating complex orbital mechanics and closed-loop agriculture into accessible browser games (Lunar farming, Brute Bowl).

Momentum & Output: The 2026 Data Pulse

997 contributions in the last year.





Dr. Richard Barker is not just analyzing disparate space data; he is coding the unified operating system for human survival beyond Earth.